
Allen Huang

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Experience

Senior Software Engineer	Klaviyo	Jul 2024 - Dec 2025
• Built the asynchronous flow-recommendation pipeline, fanning work out across 5+ partner-domain microservices via Pulsar/Kafka; modeled generation as a finite-state machine in MySQL and cached expensive compute in Redis, reducing new-customer subsequent flow generations by 66%. • Engineered the flow-classification backend that queues requests through Pulsar to the Data Science platform. • Participated in the on-call rotation for Flows; resolved P0/P1 incidents, debugged production issues in pairs, and built reusable CLI tooling that detected and repaired invalid state, eliminating a class of manual cleanup. • Collaborated with Data Science on "FlowsAI" recommendations, combining LLM output with serialized flow graphs to generate recommended automations.		
Software Engineer	Klaviyo	May 2022 - Jul 2024
• Architected the "Encoded Flows" schema-transformation system to decouple internal storage from the public contract: designed a separate serialized nodes-and-edges graph format that deliberately hides storage internals from the API. Owned the majority of the design and implementation in collaboration with my manager and the APIs team, with a comprehensive test suite which became the foundation for the developer CRUD APIs, flow classification, and recommendation generation. • Distributed transformation work across consumers via Pulsar so independent tasks could be processed asynchronously, decoupling throughput from single-threaded transformation logic. • Built the core developer flow create/read APIs on top of the Encoded Flows format, now consumed by third-party integrations at 100+ RPS. • Maintained supporting infrastructure on AWS with Terraform and Kubernetes. • Owned privacy/data-deletion compliance for the Flows domain: designed the cascading hard-delete pipeline that purges a customer's Flow object and 20+ dependent entities while preserving referential integrity across services. • Modernized Backbone.js analytics pages to React/Redux in partnership with Design/Product.		
Software Engineer Intern	PCI	Sep 2021 - Dec 2021
• Migrated the application from a GCP Kubernetes cluster to AWS and stood up a CI/CD pipeline (Jenkins, Trivy, Anchore, Harbor) with container image scanning.		
Software Engineer Intern	Klaviyo	Jun 2021 - Aug 2021
• Migrated Backbone.js analytics pages to React and built Django API endpoints for Cassandra data access, with pytest/Django test coverage for local and CI/CD execution.		

Skills

Languages: Python, JavaScript, TypeScript, SQL, Bash, Nix
Backend & Messaging: Django, Kafka, Pulsar, RabbitMQ, Redis, MySQL, PostgreSQL, Clickhouse
Distributed Systems: event-driven architecture, async fan-out across services, finite-state-machine orchestration, decoupling internal data models from public API contracts
Infrastructure & Cloud: AWS, Kubernetes, Terraform, Docker, ArgoCD, Buildkite, CI/CD, Git
Practices & Observability: pair programming, code review, on-call/incident response, test-driven development (pytest, Playwright), Grafana, Sentry, Splunk, PagerDuty
Languages: English (native), Cantonese Chinese (conversational)

Education

B.S. in Computer Science, Oregon State University -- Corvallis, OR	GPA: 3.95/4.0
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Awards & Patents

Multiple Category Wins, Klaviyo Hackathon	2024
Third Place, Klaviyo Hackathon	2023

Projects

Wikiquiz Crystal, Svelte, Python, FastAPI, Redis, Terraform, Docker, spaCy, BERT, ONNX	2021
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